

ANGEL OZA

Information and Communication Technology

Email: angeloza.work@gmail.com

Mobile: +91 9978844231

GitHub: <https://github.com/Angel-Oza>

Portfolio: angeloza.portfolio.io

Linkedin: linkedin.com/in/angel-oza

SUMMARY

ICT undergraduate with experience in embedded systems and hardware prototyping, with hands-on project experience in microcontroller-based control, sensor integration, LED/display systems, and wireless communication and building practical projects. Have taken up leadership roles in student organizations, managing teams and organizing technical events. Comfortable with problem-solving, working in teams, and learning new tools as required

EDUCATION

MARWADI UNIVERSITY

Bachelor of Technology - [Information and Communication Technology](#)

Aug 2023 -
May 2027

- Coursework: Data Structures and Algorithms, Object-oriented Programming, Database Management Systems, Operating Systems, Computer Networks, and Software engineering
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PROJECTS

1. Sign Speak — Speech-to-Sign Language & Gesture Recognition System

- Developed a real-time assistive communication system that converts speech into sign language gestures and recognises hand gestures into text/speech using Python.
- Implemented speech recognition, webcam-based gesture detection, and multilingual support using OpenCV, SpeechRecognition, NumPy, and machine learning techniques.
- Designed an interactive browser-based interface for real-time communication support for deaf and mute individuals.

2. SurakshaLink

- Developed an offline emergency communication system using ESP32 and LoRa.
 - Implemented structured wireless communication with alert prioritization.
 - Integrated acknowledgment handling for reliable node-to-node messaging.
 - Designed a real-time dashboard for communication monitoring and logs.
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RESEARCH EXPERIENCE

Dhirubhai Ambani University (Ongoing)

- Designing a **Digital Twin** of a coal mine to simulate autonomous fleet coordination, **IoT-based sensor monitoring**, and battery-powered **mobile Wi-Fi infrastructure**.
- Modeling dynamic **loading-hauling truck coordination**, incorporating resource depletion, shifting loading points, and adaptive route planning under real-time operational constraints.
- Building a **Python-based simulation** with animated visualizations and performance analytics to evaluate fleet synchronization, communication coverage, and operational efficiency.

SKILLS

Programming & Software Development

- Languages: C++, Python, Java, Kotlin
- Web Technologies: ReactJS, HTML, CSS, JavaScript

Machine Learning (Supporting)

- Frameworks: PyTorch, TensorFlow, RDK OS
- Libraries: Pandas, NumPy, Matplotlib

Hardware & Embedded (Supporting)

- Computer Architecture, Digital Systems
 - Embedded Systems: Arduino, Raspberry Pi, ESP8266/ESP32, Zigbee, D-Robotics RDK X5
 - Circuit Design & Simulation: PCB Design, Logic Design
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LEADERSHIP & STEM ACTIVITIES

- **Chair - IEEE MEFGI Student Branch (2026):** Led student operations, coordinated technical events, and collaborated with industry professionals.
- **IEEE WIE MEFGI Student Branch - Web Master :** Led the digital initiatives of the IEEE WIE student branch as a core member, managing the social website and amplifying IEEE WIEs outreach.
- **Convenor - Circuitology Club (2024 - '25):** Built a 110+ member technical community, secured USD 3,500+ in funding, and organized 30+ technical sessions and STEM outreach programs.
- **Marketing Lead - Google Developer Group On-Campus MU:** As Marketing Lead at GDG on Campus, strategized and executed the chapters branding and communication initiatives, leading design, content, and outreach efforts to build a strong tech-driven community on campus.